

Adhiraj Sengar

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TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, Dart

Deep Learning Frameworks: TensorFlow, PyTorch, Hugging Face Transformers

Libraries & Tools: NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV, Git

PROJECTS

AI-Powered Voice Phishing Detection

June 2024 – August 2024

Speech Recognition and Anomaly Detection

Python, Librosa, PyDub

- Built an offline system that transcribes call audio with Vosk and scores scam risk from 0 to 100 in 3-second streaming windows, using speech-to-text, a 119-feature acoustic IsolationForest and rule-based scam-phrase detection
- Tested on 300 held-out synthetic calls; reached F1 0.91 with text rules and 0 false alarms on legitimate calls
- Packaged with a CLI, Streamlit dashboard, 34 tests and multi-OS CI; 3 s detection latency, fully local

Hybrid Stock Market Prediction Model

June 2025 – September 2025

Deep Learning, Time-Series Forecasting & Meta-Ensembling

Python, PyTorch, Scikit-Learn

- Engineered a Stacking Meta-Ensemble combining XGBoost, Random Forest, BiLSTM, and Time-Series Transformers
- Integrated Yahoo Finance API to enrich target stock models with macro-economic indicators.
- Developed a non-linear XGBoost meta-learner dynamically weighting sub-models based on real-time market volatility
- Automated Bayesian Hyperparameter Optimization using Optuna to tune model architectures and loss functions

Narada — Offline P2P Secure Messaging

May 2026 – August 2026

Distributed Systems, Cryptography, BLE, Wi-Fi Aware

Kotlin, Flutter, Dart

- Developed a decentralized messaging application enabling serverless communication through BLE mesh, Wi-Fi Aware, Wi-Fi Direct, and optional internet relays
- Implemented end-to-end encryption using Signal Double Ratchet with hybrid key exchange for post-quantum protection
- Built store-and-forward mesh routing with packet deduplication, acknowledgements, persistent offline outbox, and multi-hop message delivery

EDUCATION

Vellore Institute of Technology

Bhopal, Madhya Pradesh

B.Tech in Computer Science and Engineering

Sep 2023 – Jun 2027

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming

CERTIFICATIONS

Google IT Support Professional Certificate

Google (Coursera)

Skills: Systems Administration, OS Management, IT Infrastructure, Troubleshooting

2026

Introduction to Machine Learning

NPTEL

Skills: Machine Learning, Regression Models, Classification Algorithms, PCA

2024

The Bits and Bytes of Computer Networking

Google (Coursera)

Skills: Networking, Network Troubleshooting, TCP/IP, DNS, Routing Protocols

2025